

A random walk through the galaxies with digressions through Chicago and Paris

J Fry, U Florida, Emeritus, Northeastern U, part time

The Random Universe: Andrew Jaffe at 60

30 June 2026











AN ESTIMATE OF Ω_m WITHOUT CONVENTIONAL PRIORS

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ABSTRACT

Using mean relative peculiar velocity measurements for pairs of galaxies, we estimate the cosmological density parameter Ω_m and the amplitude of density fluctuations σ_8 . Our results suggest that our statistic is a robust and reproducible measure of the mean pairwise velocity and thereby the Ω_m parameter. We get $\Omega_m = 0.30^{+0.17}_{-0.07}$ and $\sigma_8 = 1.13^{+0.22}_{-0.23}$. These estimates do not depend on prior assumptions on the adiabaticity of the initial density fluctuations, the ionization history, or the values of other cosmological parameters.

Subject headings: cosmological parameters — cosmology: observations — cosmology: theory — distance scale — galaxies: distances and redshifts — large-scale structure of universe

1. INTRODUCTION

In this Letter, we report the culmination of a program to study cosmic flows. In a series of recent papers, we introduced a new dynamical estimator of the Ω_m parameter, the dimensionless density of the nonrelativistic matter in the universe. We use the so-called streaming velocity, or the mean relative peculiar velocity for galaxy pairs, $v_{12}(r)$, where r is the pair separation (Peebles 1980, p. 170). It is measured directly from peculiar velocity surveys, without the noise-generating spatial differentiation, used in reconstruction schemes such as POTENT (see Courteau et al. 2000 and references therein). In the first paper of the series (Juszkiewicz, Springel, & Durrer 1999), we derived an equation relating $v_{12}(r)$ to Ω_m and the two-point correlation function of mass density fluctuations, $\xi(r)$. Then, we showed that v_{12} and Ω_m can be estimated from mock velocity surveys (Ferreira et al. 1999) and finally, from real data: the Mark III survey (Juszkiewicz et al. 2000). Whenever a new statistic is introduced, it is of particular importance that it passes the test of reproducibility. Our Mark III results pass these tests: the $v_{12}(r)$ measurements are independent of the galaxy morphology and the distance indicator.

In this Letter, we extend our analysis to three new surveys,

with the aim of testing reproducibility on a larger sample and, in case of a positive outcome, improving on the accuracy of our earlier measurements of Ω_m and σ_8 , the rms mass density contrast in a sphere of radius of $8 h^{-1}$ Mpc, where h is the usual Hubble parameter, H_0 , expressed in units of $100 \text{ km s}^{-1} \text{ Mpc}^{-1}$. In our notation, the symbol σ_8 always refers to matter density, while σ_8^{PSCz} refers to the number density of Point-Source Catalog Redshift (PSCz) survey galaxies.

Unlike our analysis, other estimators of cosmological parameters are often degenerate, hence σ_8 and Ω_m cannot be extracted without making additional Bayesian prior assumptions, which we call conventional priors: a particular choice of values for h , the baryon and vacuum densities, Ω_b and Ω_Λ , the character of the primeval inhomogeneities (adiabaticity, spectral slope, t/s ratio), the ionization history, etc. (Bridle et al. 2003). The estimates of Ω_m and σ_8 presented here *do not* depend on conventional priors. The only prior assumption that we make is that up to σ_8 , the PSCz estimate of $\xi(r)$ describes the mass correlation function. We test this assumption by comparing the predicted $v_{12}(r)$ to direct observations. We also check how robust our approach is by replacing the PSCz estimate of $\xi(r)$ with an automatic plate measuring (APM) estimate and two other pure power-law toy models.

2. THE PAIRWISE MOTIONS AND GALAXY CLUSTERING

The approximate solution of the pair conservation equation derived by Juszkiewicz et al. (1999) is given by

$$v_{12}(r) = -\frac{2}{3} H_0 r \Omega_m^{0.6} \bar{\xi}(r) [1 + \bar{\omega}(r)] \quad (1)$$

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Weakly nonlinear dynamics and the σ_8 parameter

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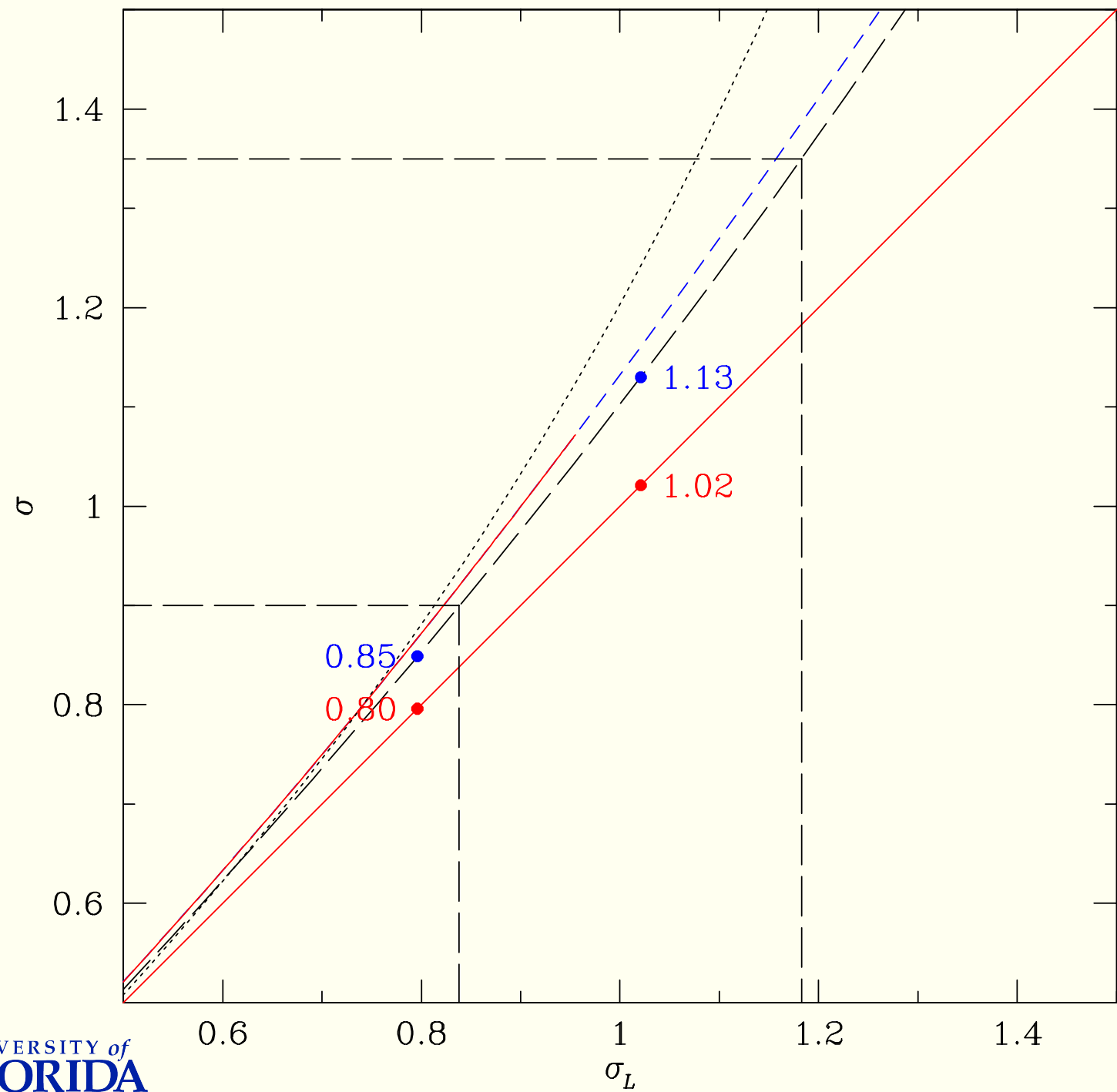
Revised January 19, 2010

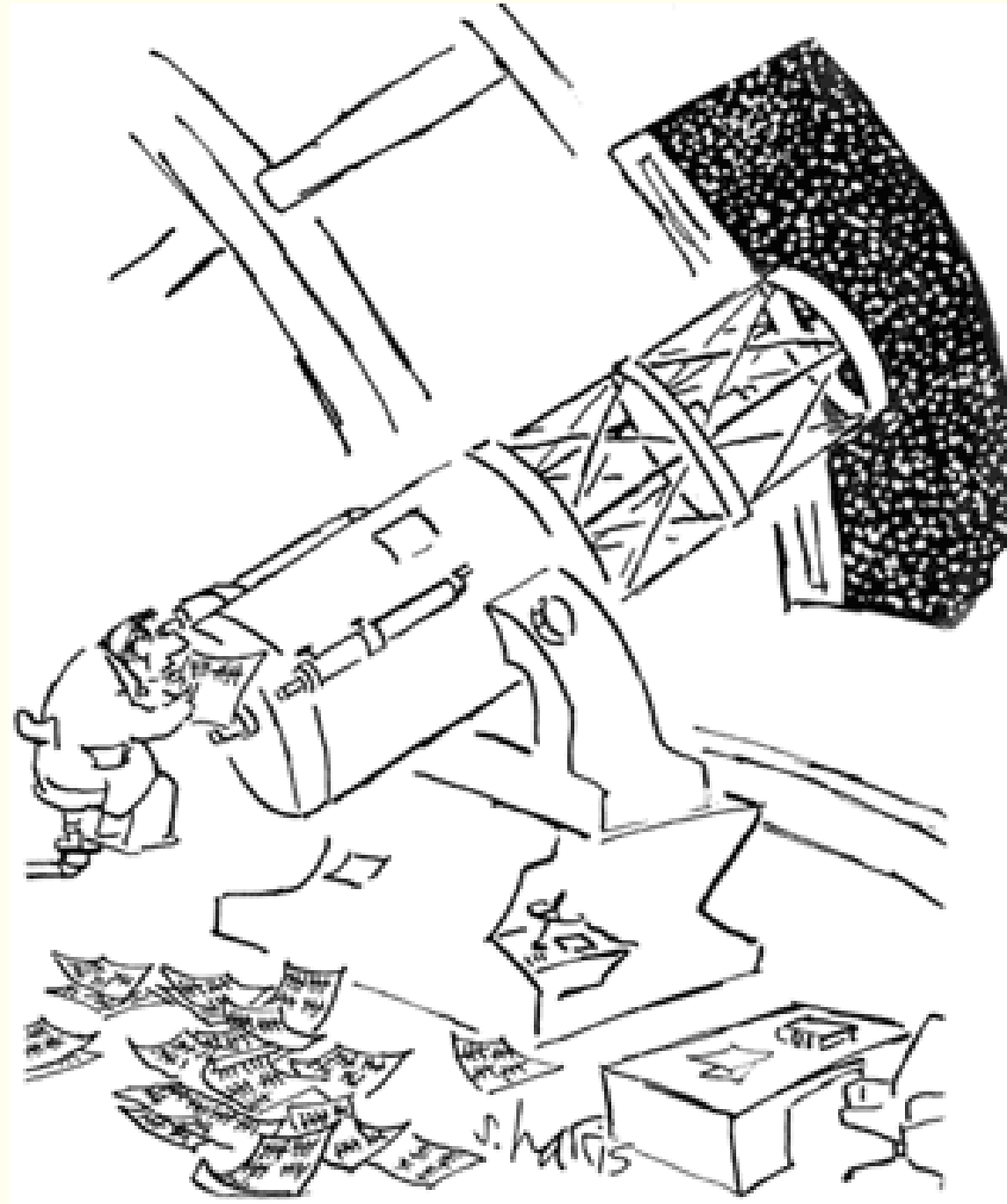
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Abstract. The amplitude of cosmological density fluctuations, σ_8 , has been studied and estimated by analysing many cosmological observations. The values of the estimates vary considerably between the various probes. However, different estimators probe the value of σ_8 in different cosmological scales and do not take into account the nonlinear evolution of the parameter at late times. We show that estimates of the amplitude of cosmological density

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Halo Model

Objects or events come in clusters, or halos: $N = \sum_{i=1}^{N_h} N_i$

Has been applied in: high energy collisions (hadron multiplicities), photon shot noise statistics (cathodeluminescence radiation from a pulsed electron beam), neurophysiology of vision (nerve impulses), geophysics (earthquake events), hydrology (rainfall totals), species intensity (insect larvae in fields, pine trees in natural forests, Minke whales), diamond deposits, and (doubly) 1990 storm damage in the Fontainebleau forest

And: galaxy count statistics

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B. M. BROWN AND ANN COWLING

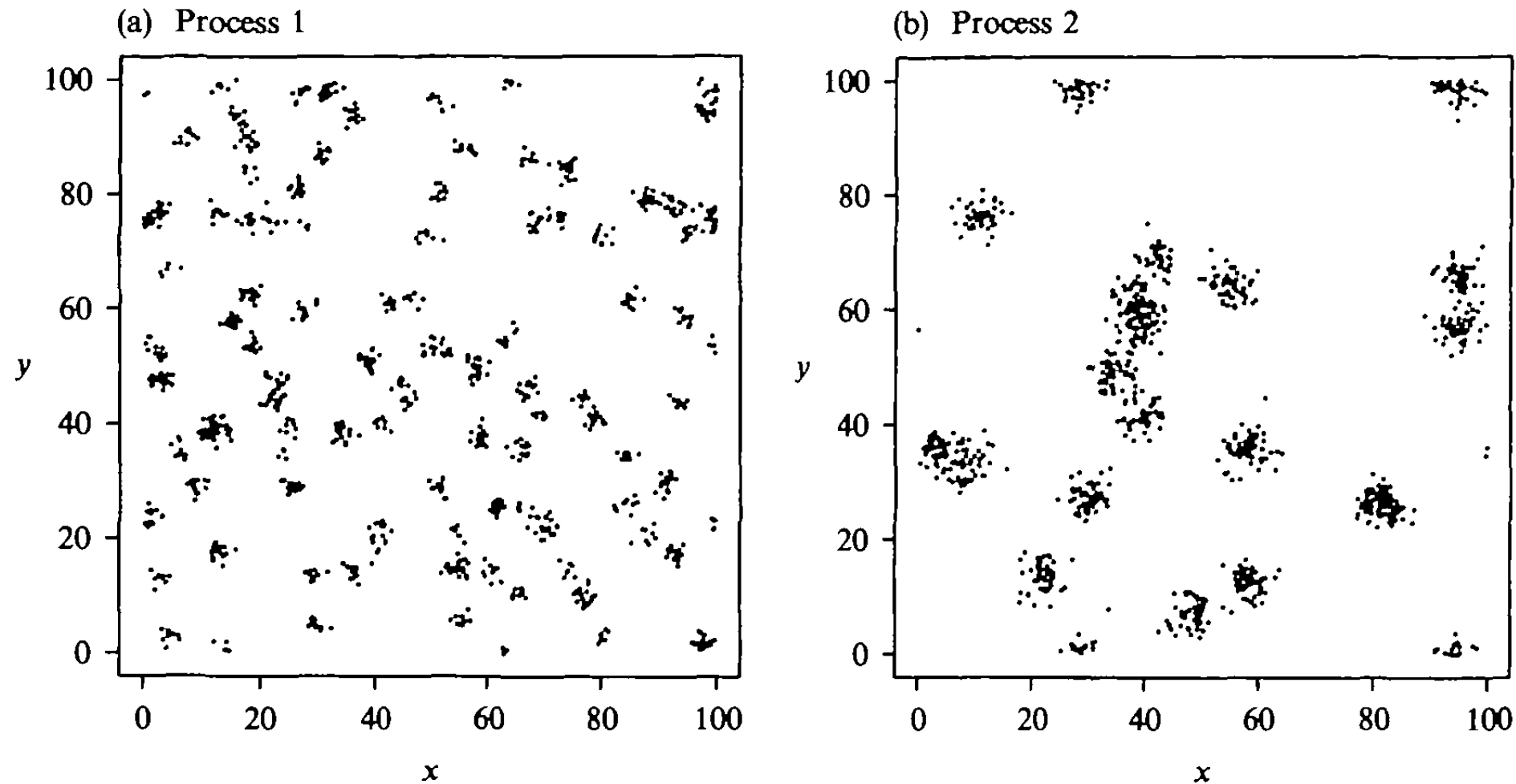


Fig. 1. Realisations of the processes used in the simulation study: (a) process 1, $(\lambda, \mu, \rho) = (0.01, 10, 1.0)$; (b) process 2, $(\lambda, \mu, \rho) = (0.002, 50, 2.24)$.

Construction d'un modèle de mortalité "Tempête"
basé sur la répartition spatiale des chablis en peuplement.
Application à la réserve intégrale de la Tillaie.

François Goreaud (LISC, CEMAGREF)

1332. Exemple des modèles construits pour les tempêtes de 1967 et 1990 :

L'analyse de la structure spatiale des arbres morts pendant les deux tempêtes de 1967 et 1990 (Goreaud et al., 2000) nous a permis de confirmer le caractère agrégé de la mortalité (voir figure en annexe 3). A partir de ces observations, nous avons pu construire des modèles simples de mortalité par tempête, correspondant à cette structure spatiale, en utilisant des processus ponctuels. Pour la tempête de 1967, nous avons utilisé un processus de Neyman-Scott simple (Tomppo, 1986), et pour la tempête de 1990, nous avons construit un processus ponctuel doublement agrégé (figure 3), en composant deux processus de Neyman-Scott.

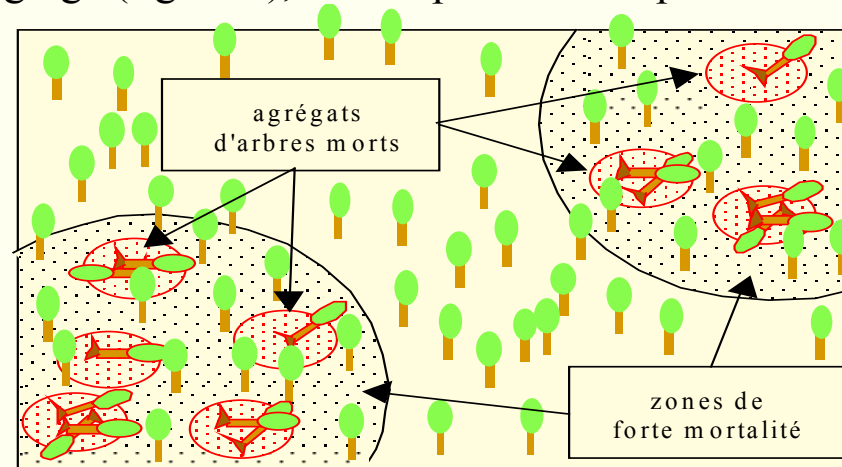


Figure 3 : Principe du modèle de mortalité par tempête doublement agrégé, construit pour rendre compte des dégâts dus à la tempête de 1990.

Non-conditional and conditional simulation of a spatial point process

J. Caers, J. Gelders, A. Vervoort, and L. Rombouts

Geostatistics Wollongong '96, vol. 1, p.270 (Dordrecht: Kluwer, 1997)

SPATIAL POINT PROCESS

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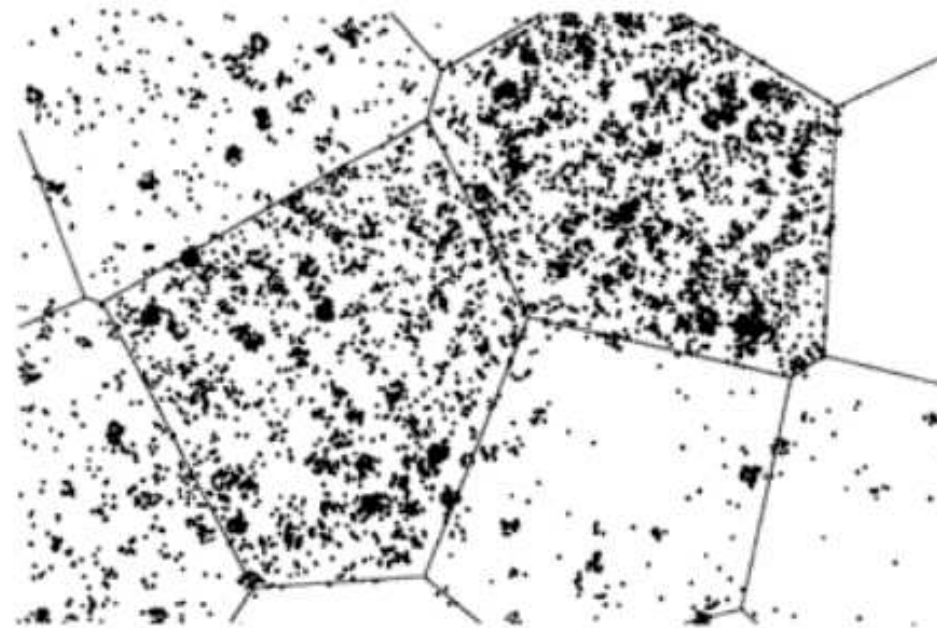
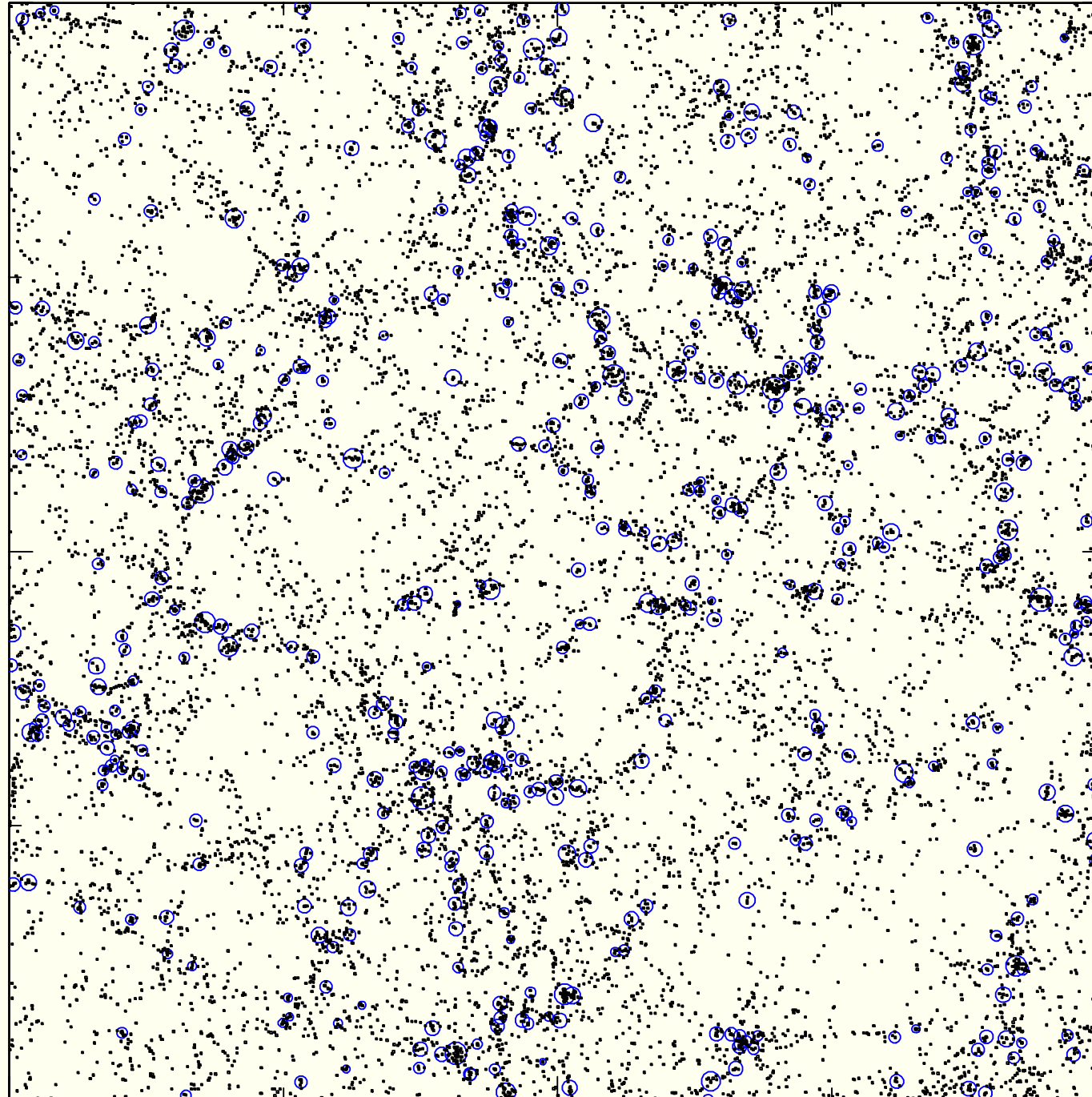


Figure 2. Non-conditional simulation of a diamond deposit



Generating Functions

$$M(t) = \sum_{k=0}^{\infty} \frac{1}{k!} \bar{\mu}_k t^k = \langle e^{tx} \rangle \qquad K(t) = \sum_{k=0}^{\infty} \frac{1}{k!} \bar{\xi}_k t^k = \log M(t)$$

$$G(z) = \sum_{n=0}^{\infty} p_n z^n = \langle z^n \rangle$$

$$p_n = \frac{1}{n!} \frac{d^n}{dz^n} G(z) \Big|_{z=0} \qquad \langle N^{[k]} \rangle = \frac{d^k}{dz^k} G(z) \Big|_{z=1}$$

$$\langle N^{[k]} \rangle = \frac{d^k}{dt^k} M_c(t) \Big|_{t=0} \qquad p_n = \frac{1}{n!} \frac{d^n}{dt^n} M_c(t) \Big|_{t=-1}$$

$$M_{\text{discrete}}(t) = M_{\text{continuous}}(e^t - 1)$$

$$G(z) = M(z - 1)$$

Halo Model Generating Function

Point Cluster Limit (iid) Szapudi & Szalay 1993

$$\begin{aligned} G(z) &= \langle z^N \rangle = \langle \langle z^{N_1 + \dots + N_{N_h}} \rangle \rangle \\ &= \langle \langle z^{N_1} \rangle \dots \langle z^{N_{N_h}} \rangle \rangle = \langle [g_i(z)]^{N_h} \rangle = g_h[g_i(z)] \end{aligned}$$

$$\begin{aligned} K(t) &= \log[M(t)] = \log[G(t+1)] = \log(g_h[g_i(t+1)]) \\ &= \log(g_h[M_i(t)]) = \log(M_h[M_i(t) - 1]) = K_h[M_i(t) - 1] \end{aligned}$$

Generating Function Results for Cell Count Statistics

$$\bar{N} = \bar{N}_h \bar{N}_i$$

$$\bar{\xi}_2 = \bar{\xi}_{2,h} + \frac{\bar{\mu}_{2,i}}{\bar{N}_h}$$

$$\bar{\xi}_3 = \bar{\xi}_{3,h} + \frac{3 \bar{\mu}_{2,i} \bar{\xi}_{2,h}}{\bar{N}_h} + \frac{\bar{\mu}_{3,i}}{\bar{N}_h^2}$$

$$\bar{\xi}_4 = \bar{\xi}_{4,h} + \frac{6 \bar{\mu}_{2,i} \bar{\xi}_{3,h}}{\bar{N}_h} + \frac{(4 \bar{\mu}_{3,i} + 3 \bar{\mu}_{2,i}^2) \bar{\xi}_{2,h}}{\bar{N}_h^2} + \frac{\bar{\mu}_{4,i}}{\bar{N}_h^3}$$

$$\bar{\xi}_5 = \bar{\xi}_{5,h} + \frac{10 \bar{\mu}_{2,i} \bar{\xi}_{4,h}}{\bar{N}_h} + \frac{(10 \bar{\mu}_{3,i} + 15 \bar{\mu}_{2,i}^2) \bar{\xi}_{3,h}}{\bar{N}_h^2} + \frac{(10 \bar{\mu}_{2,i} \bar{\mu}_{3,i} + 5 \bar{\mu}_{4,i}) \bar{\xi}_{2,h}}{\bar{N}_h^3} + \frac{\bar{\mu}_{5,i}}{\bar{N}_h^4}$$

$$\begin{aligned}
\bar{\xi}_9 = & \bar{\xi}_{9,h} + \frac{36 \bar{\mu}_2 \bar{\xi}_{8,h}}{\bar{N}_h} + \frac{(378 \bar{\mu}_2^2 + 84 \bar{\mu}_3) \bar{\xi}_{7,h}}{\bar{N}_h^2} + \frac{(1260 \bar{\mu}_2^3 + 1260 \bar{\mu}_2 \bar{\mu}_3 + 126 \bar{\mu}_4) \bar{\xi}_{6,h}}{\bar{N}_h^3} \\
& + \frac{(945 \bar{\mu}_2^4 + 3780 \bar{\mu}_2^2 \bar{\mu}_3 + 840 \bar{\mu}_3^2 + 1260 \bar{\mu}_2 \bar{\mu}_3 + 126 \bar{\mu}_5) \bar{\xi}_{5,h}}{\bar{N}_h^4} \\
& + \frac{(1260 \bar{\mu}_2^3 \bar{\mu}_3 + 2520 \bar{\mu}_2 \bar{\mu}_3^2 + 1890 \bar{\mu}_2^2 \bar{\mu}_4 + 1260 \bar{\mu}_3 \bar{\mu}_4 + 756 \bar{\mu}_2 \bar{\mu}_5 + 84 \bar{\mu}_6) \bar{\xi}_{4,h}}{\bar{N}_h^5} \\
& + \frac{(280 \bar{\mu}_3^3 + 1260 \bar{\mu}_2 \bar{\mu}_3 \bar{\mu}_4 + 315 \bar{\mu}_4^2 + 378 \bar{\mu}_2^2 \bar{\mu}_5 + 504 \bar{\mu}_3 \bar{\mu}_5 + 252 \bar{\mu}_2 \bar{\mu}_6 + 36 \bar{\mu}_7) \bar{\xi}_{3,h}}{\bar{N}_h^6} \\
& + \frac{(126 \bar{\mu}_4 \bar{\mu}_5 + 84 \bar{\mu}_3 \bar{\mu}_6 + 36 \bar{\mu}_2 \bar{\mu}_7 + 9 \bar{\mu}_8) \bar{\xi}_{2,h}}{\bar{N}_h^7} + \frac{\bar{\mu}_{9,i}}{\bar{N}_h^8}
\end{aligned}$$

Voids and Hierarchical Scaling

Fall, Geller, Jones & White 1976; White 1979; Fry 1984

$$P_0 = G(0) = M(-1) = \exp \left[\sum_{p=1}^{\infty} \frac{(-1)^p}{p!} \bar{N}^p \bar{\xi}_p \right]$$

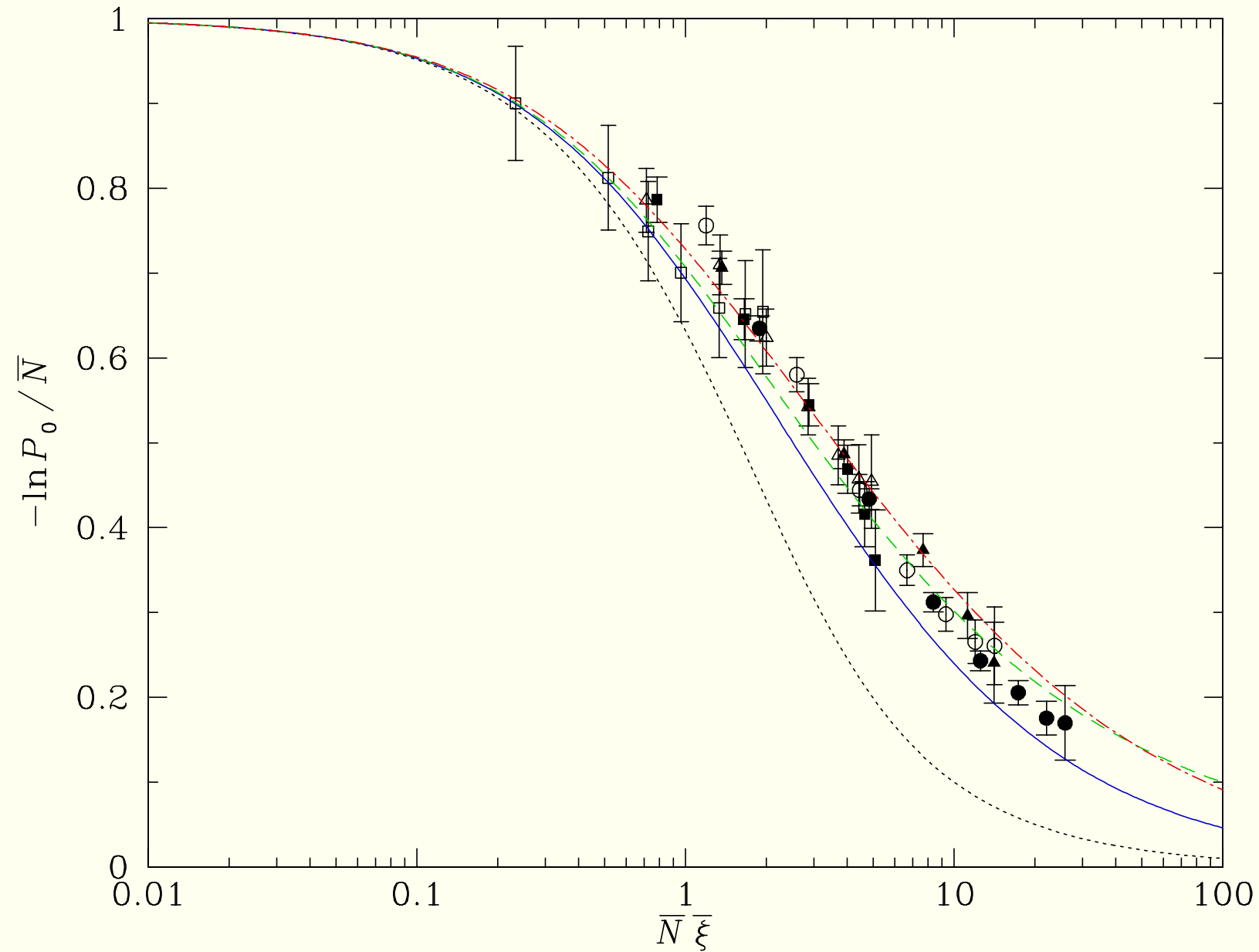
Hierarchical $\bar{\xi}_p = S_p \bar{\xi}^{p-1}$ Sharp 1981; Schaeffer 1984; Fry 1986; Balian & Schaeffer 1989

$$P_0 = \exp \left[-\bar{N} \sum_{p=1}^{\infty} \frac{(-1)^{p-1}}{p!} S_p (\bar{N} \bar{\xi})^{p-1} \right]$$

$$-\frac{\log P_0}{\bar{N}} = \chi(\bar{N} \bar{\xi}), \quad \bar{N} \bar{\xi} = \frac{\langle N^2 \rangle - \langle N \rangle^2 - \langle N \rangle}{\langle N \rangle}$$

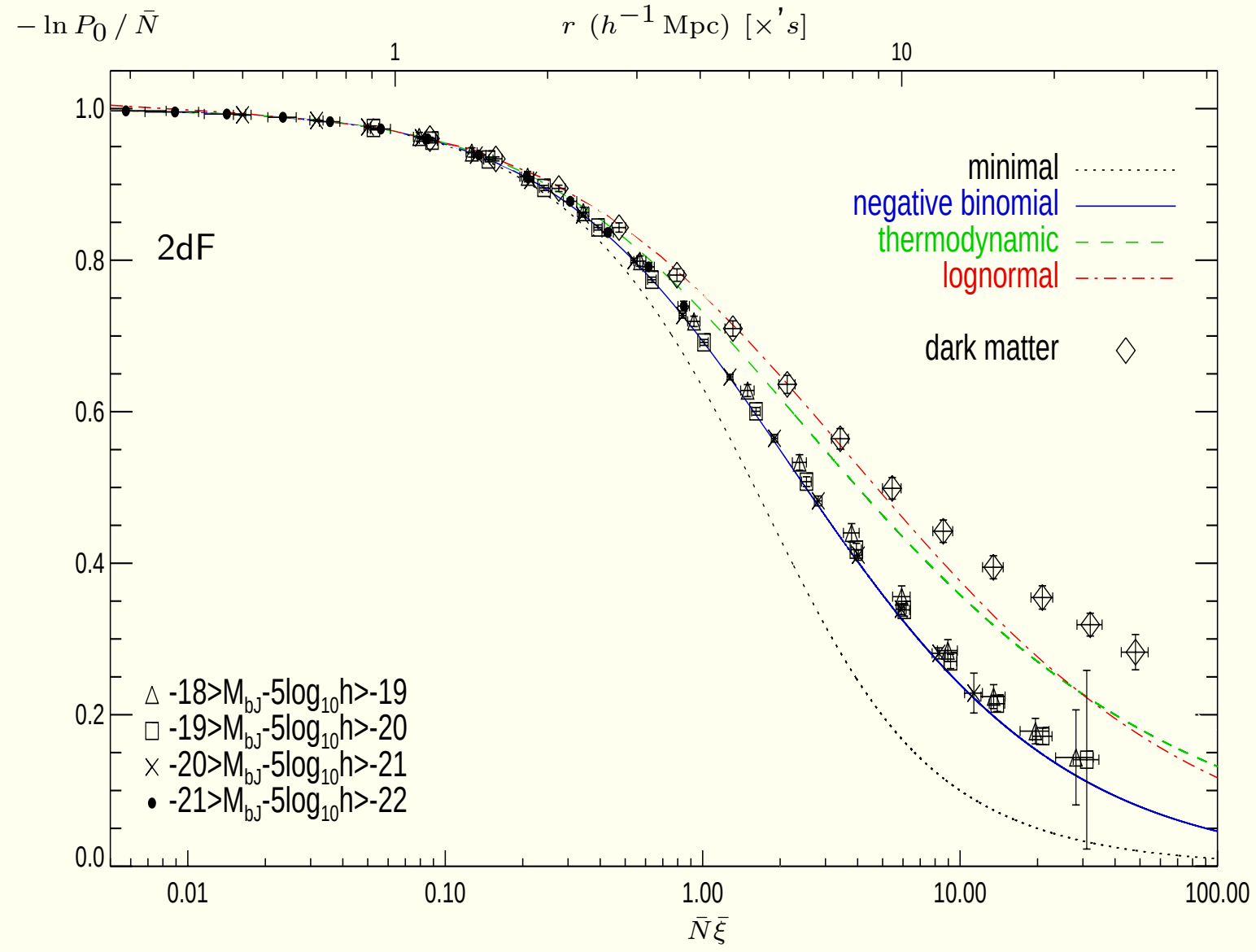
Void statistics, scaling, and the origins of large-scale structure

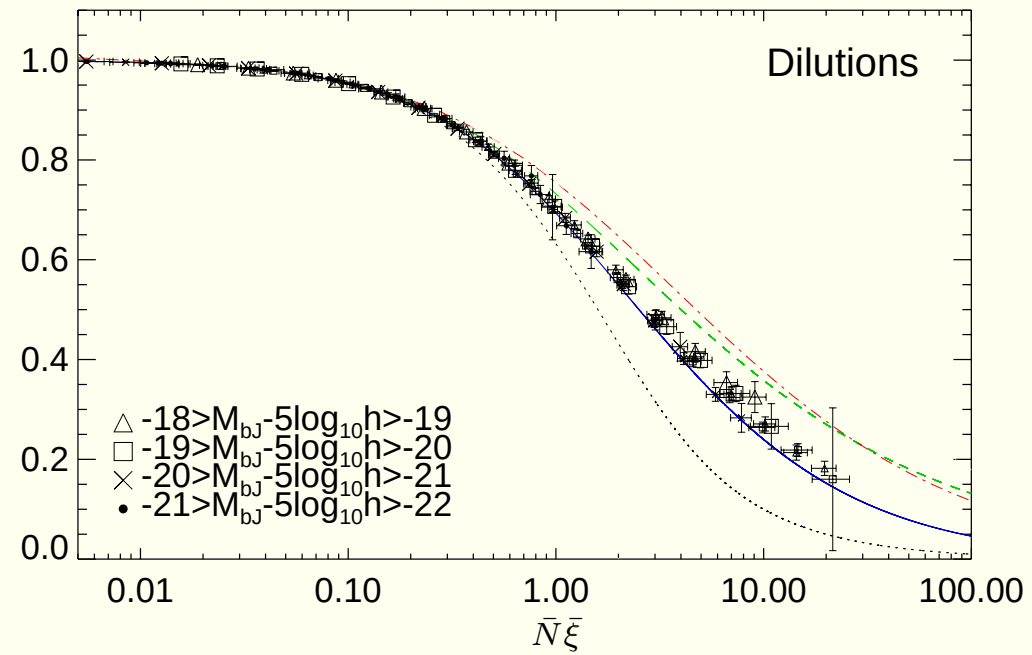
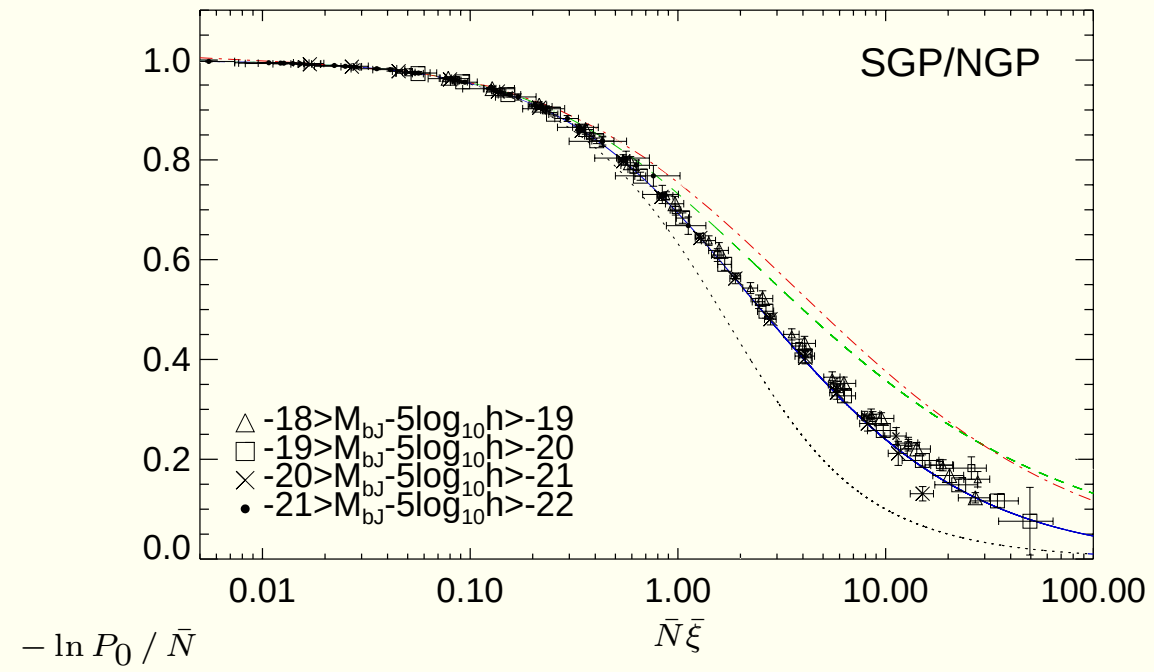
Fry, Giovanelli, Haynes, Melott, & Scherrer *ApJ* 1989

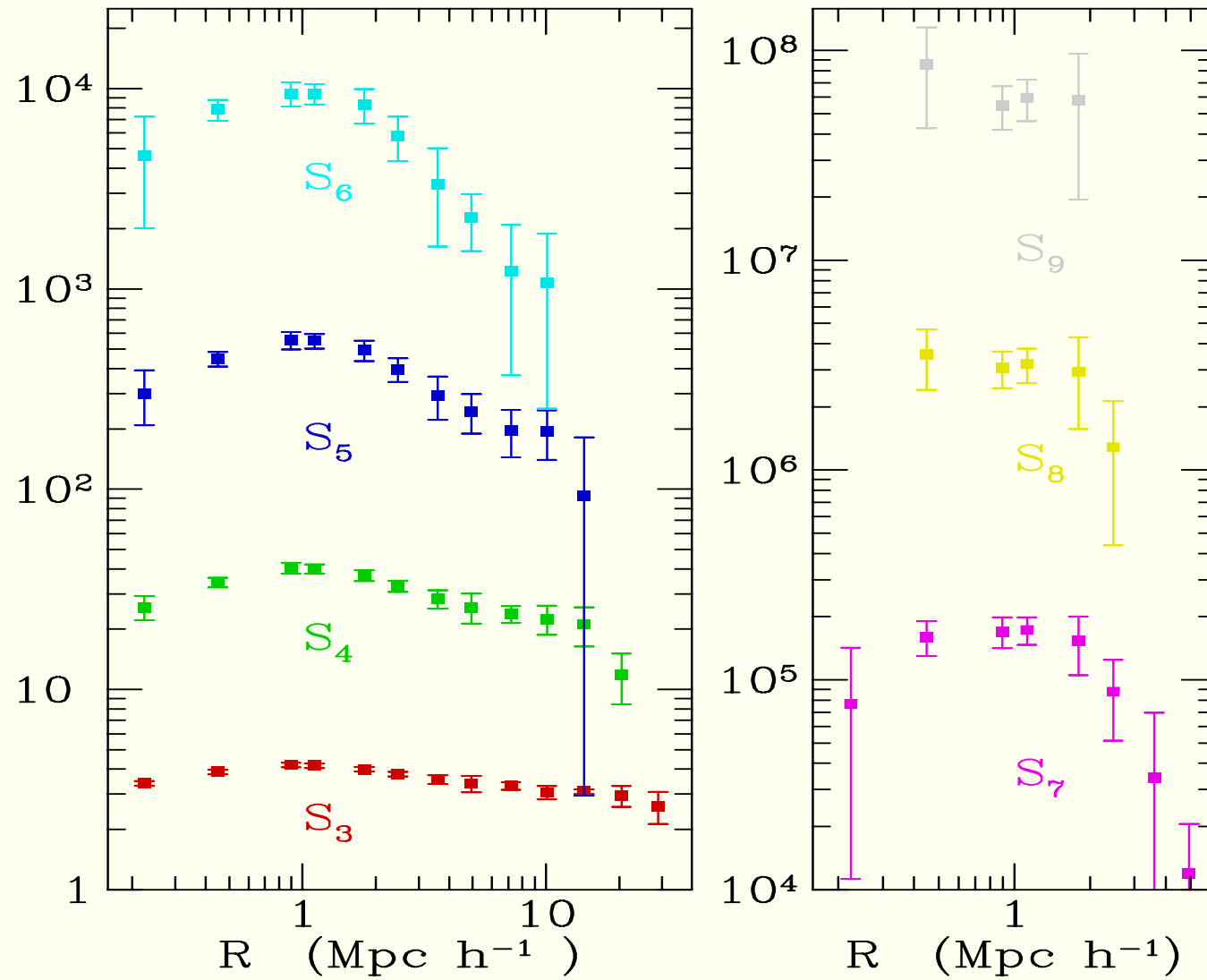


The 2dF Galaxy Redshift Survey: voids and hierarchical scaling models

Croton et al. 2004, MNRAS 352, 828







E. Gaztanaga 1994, MNRAS, 269, 915, Figure 5

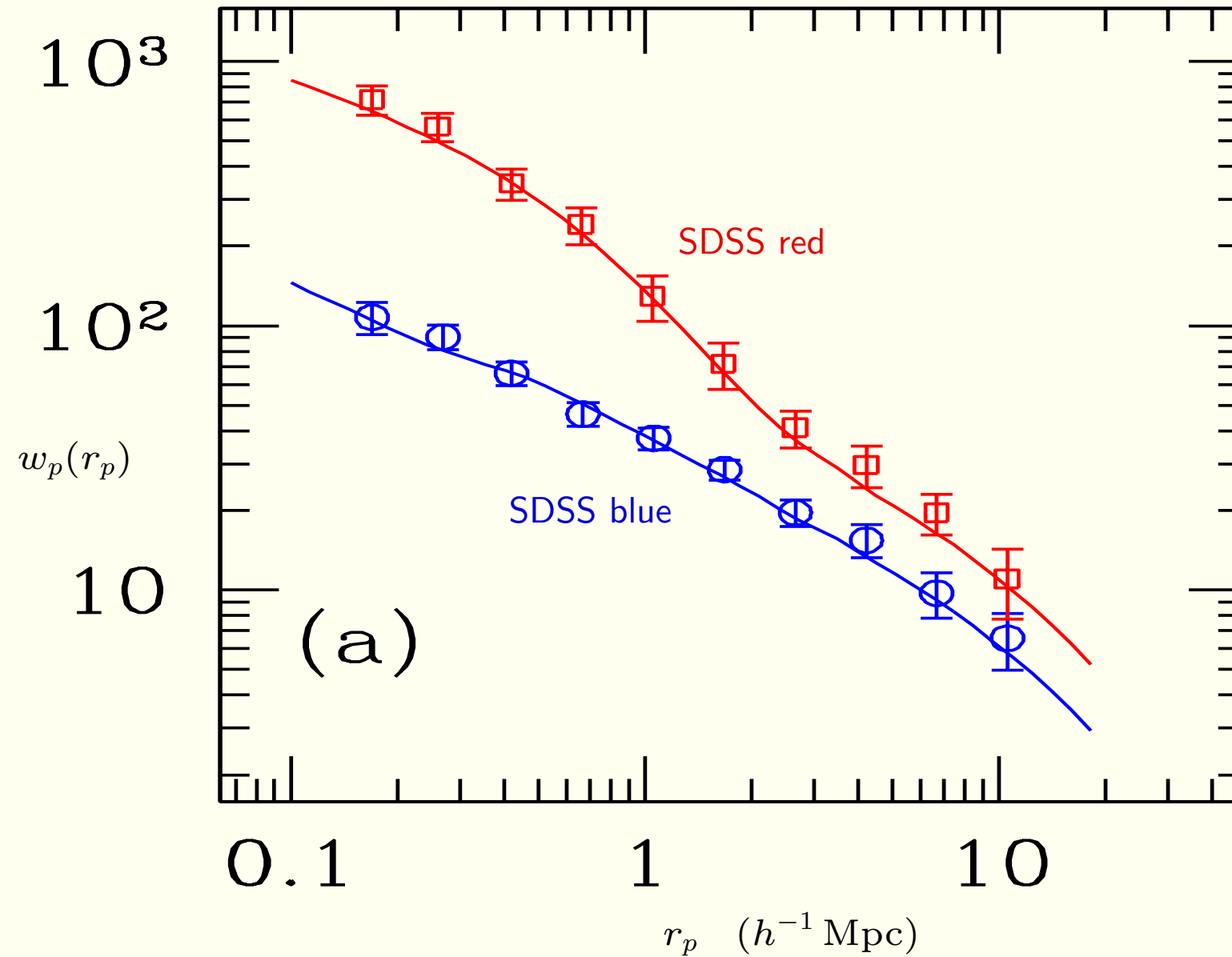
Void Scaling in the Halo Model

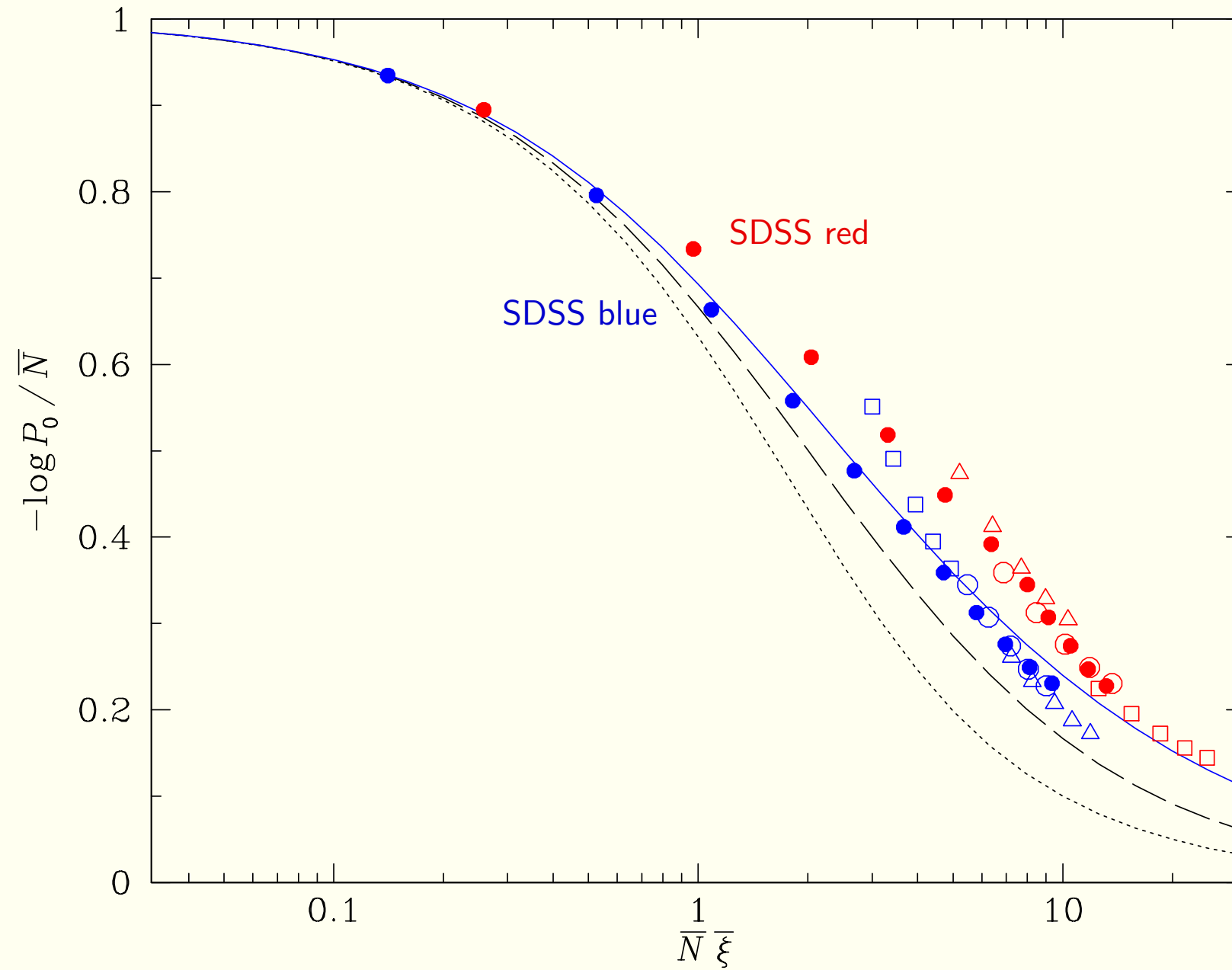
$$\bar{N}_g \bar{\xi}_g = \frac{\bar{N}_g}{\bar{N}_h} (\bar{N}_h \bar{\xi}_h + \bar{\mu}_2)$$

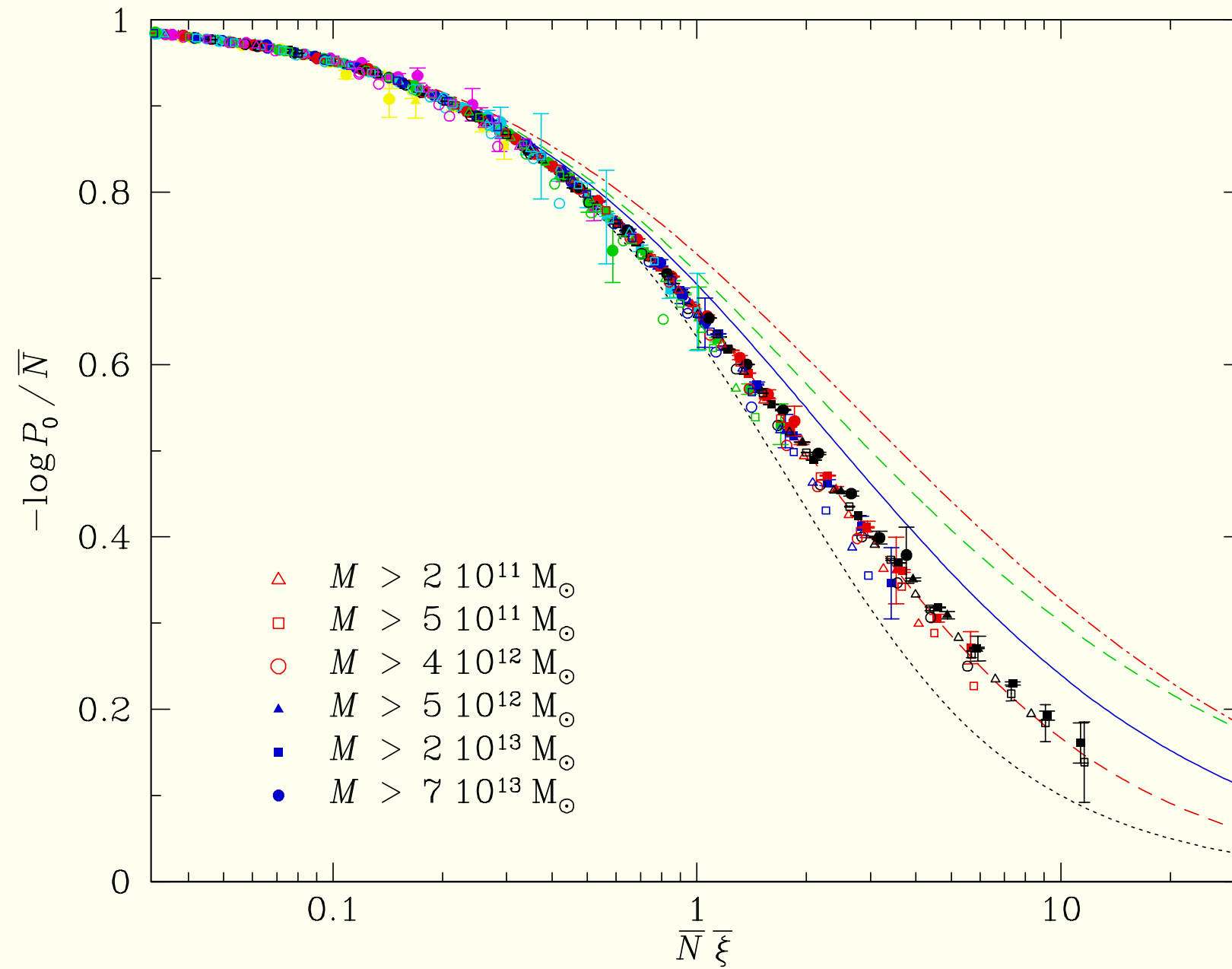
$$\bar{N}_g^2 \bar{\xi}_{3,g} = \frac{\bar{N}_g^2}{\bar{N}_h^2} [S_{3,h} (\bar{N}_h \bar{\xi}_h)^2 + 3\bar{\mu}_2 \bar{N}_h \bar{\xi}_h + \bar{\mu}_3],$$

$$\bar{N}_g^3 \bar{\xi}_{4,g} = \frac{\bar{N}_g^3}{\bar{N}_h^3} [S_{4,h} (\bar{N}_h \bar{\xi}_h)^3 + 6\bar{\mu}_2 S_{3,h} (\bar{N}_h \bar{\xi}_h)^2 + (4\bar{\mu}_3 + 3\bar{\mu}_2^2) \bar{N}_h \bar{\xi}_h + \bar{\mu}_4].$$

$$K(t) = K_h [M_i(t) - 1] = \frac{\bar{N}_g}{\bar{N}_i} \sum_{p=1}^{\infty} \frac{(-1)^p}{p!} (1 - M_i)^p S_{p,h} (\bar{N}_h \bar{\xi}_h)^{p-1}$$







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